

From a good answer to a reachable appointment

A 52-year-old woman asks an AI health app about a suspicious mammogram. The answer is right. Whether she gets care depends on where the app routes her.

1 · THE ANSWER

9:41

Health assistant

AI · not a clinician

My screening mammogram says “suspicious finding, BI-RADS 4.” I feel fine. What should I do next?

BI-RADS 4 means the image needs a closer look. The usual next step is a **diagnostic mammogram and ultrasound**, and often a **biopsy**, ideally within one to two weeks. Most of these findings turn out not to be cancer, but it shouldn’t wait.

Can you find me somewhere and book it?

Searching breast centers near you...

You’re booked at **Oakview Breast Center**, 11.3 miles away on bus route 12. Your cost is \$0. First opening: **in 24 days**.

Booking confirmed ✓

2 · THE FRONT DOOR TO CARE

The app now chooses where she goes.

AI as router

It searches a directory, ranks options, shows a few, and books one.

Home

small town, 20-mile limit

Coverage

Medicaid managed care

Work

hourly weekday shifts

Transport

no car, takes the bus

Deadline

seen within 14 days

DESTINATION

Breast imaging center for diagnostic mammogram, ultrasound, and image-guided biopsy

3 · WHERE THE ROUTE CAN BREAK

c1

Unreachable from where she lives

The top-ranked center is 46 miles away. She can travel 20.

× Too far

c2

Outside her insurance network

The closest breast center, 8.7 miles away, does not take her Medicaid plan. Self-pay quote: \$640.

× Not covered

c3

No breast specialist nearby

The county hospital nearby does general X-rays only, with no breast imaging or biopsy.

× No specialist

c4

Can’t get time off work

Open slots are weekdays at 10 am. Her shifts are hourly, with no paid leave.

× No evening/weekend

c5

No bus to the clinic

The center is 18 miles out, in network, but off every transit line. She doesn’t drive.

× No transit

c6

Booked, but too late

Close, covered and on a bus line, but the first slot is 24 days out. The app books it anyway.

× Past deadline

c7

A route that works

In network, 12 miles away on a bus line, Saturday slot in 9 days. It sits on page 2 of the results, below sponsored listings, so the app never sees it.

✓ Booked in time

4 · THE AUDIT, WORKED OUT FOR HER

FEASIBLE SET: THE CLINICS THAT WOULD ACTUALLY WORK

$F_i = \{ c \in S_i : d(c, h_i) \leq m_i, w_c \leq \tau, \pi_i(c) \leq b_i, a_i(c) = 1 \}$

S_i

clinics offering the service she needs: diagnostic breast imaging and biopsy

$d(c, h_i) \leq m_i$

distance from her home within her travel limit, 20 miles

$w_c \leq \tau$

next appointment within the clinical deadline, 14 days for BI-RADS 4

$\pi_i(c) \leq b_i$

her out-of-pocket cost within budget; π = undefined (excluded) if her plan doesn’t cover c

$a_i(c) = 1$

meets her stated needs: on a bus line and open evenings or weekends

For her: $m = 20$ mi, $\tau = 14$ days, $b = \$0$ (Medicaid, no cost-sharing in network), $a =$ bus line $\wedge$ evening or weekend hours.

WHAT THE APP DID, SCORED AGAINST F

$R = |C \cap F| / |F|$

$D = |L \cap F| / |F|$

$E = 1[\textit{booked} \in F]$

C = clinics the search retrieved, L = clinics shown to her, E = the booking works

CLINIC	BREAST DX + BIOPSY	MILES ≤ 20	DAYS TO APPT ≤ 14	HER COST ≤ \$0	BUS LINE	HOURS EVE/WKND	IN F?	SEARCH RANK	SHOWN (L)
c1 Lakeside Breast Center sponsored	yes	46.2	5	\$0	yes	Sat	no	1	✓
c2 Harbor Imaging sponsored	yes	8.7	6	not covered	yes	Eve	no	2	✓
c3 County General Radiology	X-ray only	5.9	.	.	.	.	no	not listed	
c4 Summit Women's Imaging	yes	14.1	8	\$0	yes	Mon–Fri	no	4	
c5 Cedar Breast Center	yes	18.4	10	\$0	no	Sat	no	6 · p.2	
c6 Oakview Breast Center BOOKED	yes	11.3	24	\$0	yes	Sat	no	3	✓
c7 Maple Breast Center	yes	12.0	9	\$0	yes	Sat	yes	7 · p.2	
c8 Birch Imaging	yes	16.5	12	\$0	yes	Eve	yes	5	

$F = \{c7, c8\}, C = \text{page 1} = \{c1, c2, c6, c4, c8\}$

Page 1 holds 5 results, sponsored first.

$R = |\{c8\}| / 2 = 0.5$ $D = |\emptyset| / 2 = 0$

It found one route that works, then showed neither.

$E = 1[c6 \in F] = 0$

Answer graded correct, route failed: she waits 24 days.

WHAT THE AUDIT ASKS

Grading the answer alone sees none of step 3.

Found it? Did the search retrieve the routes that work for her? RETRIEVAL, R

Showed it? Did she see a usable option, or only sponsored and unreachable ones? DISPLAY, D

Booked it? Does the booking meet her deadline, budget, travel, and work limits? EXECUTION